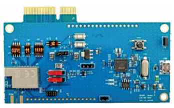
ternet of Things (IoT) devices by enabling the connection of different smart IoT devices with each other, irrespective of the manufacturer. DA16200



is both robust and cryptographically secure, making it suitable for both residential and industrial applications. It is fully integrated into Renesas' E2Studio for easy system development using Renesas MCU, connectivity, and other devices. The kit will simplify the development process, reduce the development cost, and enable faster time-to-market for IoT devices.

Renesas

https://www.renesas.com

Turnkey optical module

Teledyne e2v's Optimom 1.5M is a 1.5-megapixel turnkey optical module added to the company's Optimom series. It features a CMOS image sensor capable of capturing images at a resolution of 1920 × 800 pixels. The Optimom 1.5M comprises a proprietary image sensor, a tiny 25mm square

board with lens mount, and an optional lens in various options. The module features a native MIPI CSI-2 protocol and comes with a complete development kit, which includes an adaptor board, cabling, and Linux drivers for instant integration with MIPI based processing units, such as NVIDIA Jetson or NXP i.MX solutions. The new module can help designers to reduce development time and cost for embedded vision and AI vision solutions. The resolution of the sensor makes it suitable for applications such as laboratory equipment, handheld scanners, auto ID systems, and use as a drone camera.

Teledyne e2v

https://www.teledyne-e2v.com

Vision processors

Texas Instruments (TI) has released a new family of six Arm Cortex based vision processors that allow designers to add enhanced vision and artificial intelligence (AI) processing. This new family includes the AM62A, AM68A, and AM69A processors that can bring intelligence from the cloud to the real world by eliminating cost and design complexity barriers when implement-

ing vision processing and deep learning capabilities in low-power edge AI applications. These processors feature a system-on-a-chip (SoC) architecture that includes extensive integration. Integrated components include Arm Cortex-A53 or Cortex-A72 central processing units, a third-generation TI image signal processor, internal memory, interfaces, and hardware accelerators that deliver from 1 to 32 tera operations per second of AI processing for deep learning algorithms. They are supported by open source evaluation and model development tools and common software that is programmable through industry-standard application programming interfaces, frameworks, and models. The ARM based vision processors can reduce cost and complexity of development. These can be used in a wide range of applications, such as video doorbells, machine vision, and autonomous mobile robots.

Texas Instruments

https://www.ti.com

Automotive wireless microcontroller

Onsemi NCV-RSL15 is an ultra-low-power automotive-grade wireless microcontroller with Bluetooth Low

Energy connectivity. It is the industry's lowest power, secure microcontroller featuring the proprietary smart sense power mode. It can enable wireless connectivity, which can help vehicle manufacturers to reduce the cost and weight of the system by eliminating excess cabling required for sensors and in-vehicle communication. The NCV-RSL15 offers enhanced security with Arm CryptoCell featuring hardware based root-of-trust secure boot, many user-accessible hardware-accelerated cryptographic algorithms, and Firmware-over-the-Air capabilities to enable future firmware updates and deployment of security patches. The microcontroller addresses heightened security concerns resulting from more sensors and with that more possible attack vectors. The NCV-RSL15 will enable designers to add more sensors while reducing the wiring size, time of development, and cost of the upcoming vehicles.

Onsemi

https://www.onsemi.com

Compact standalone Wi-Fi module

u-Blox IRIS-W1 is the first compact standalone Wi-Fi module combining dual-band Wi-Fi 6, Bluetooth LE (Low Energy) 5.3, and Thread including support for Matter. The IRIS-W1 is powered by an Arm Cortex-M33

MCU that eliminates the need for an external processing unit. It comes in four variants, each enabling the user to select among various antennas and software environments. The module also enables users to select Wi-Fi channels from the 2.4 and 5GHz spectrum. It enables the engineers to create products that offer higher availability of network and throughput, less data congestion in dense environments, and lower network deployment costs. Its design makes it compatible with battery-operated devices. With its support for Matter over Thread, Wi-Fi, and Ethernet, this module provides the key technologies for intelligent buildings, smart homes, etc.

u-Blox

https://www.u-blox.com